KAVYA RANJAN SAXENA, PH.D.

Machine Learning for Audio Signal Processing | IIT Kanpur, India

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SUMMARY

I have a demonstrably strong research background in Music Information Retrieval (MIR), Domain Adaptation, and exploratory work in Speech LLMs. I have experience in designing scalable research prototypes and curating high-quality datasets. Skilled at translating complex research ideas into practical, implementable solutions that support experimentation, reproducibility, and real-world applicability. Enthusiastic about cross-disciplinary collaboration, mentoring, and advancing state-of-the-art in audio ML with a focus on measurable, practical outcomes. I have defended my Ph.D. thesis on 28th April 2026.

WORK EXPERIENCE

Principal Project Officer

IIT Madras (Bodhan AI Project)

02/2026 - 05/2026   Remote, India

- Provided research expertise as a consultant focused on enhancing oral reading fluency (ORF) among children.

Intern - Speech Research Scientist

Krutrim AI

08/2024 - 03/2025   Bangalore, India

- Explored several SOTA speech LLMs, fine-tuned them using LoRA, and conducted a comparative study of datasets for ASR tasks.
- Explored various methods used for generating synthetic data, and curated a synthetic dataset called IndicST for AST tasks (**published in ICASSPW 2025**)

Tutor

IIT Kanpur

12/2023 – 01/2024   Kanpur, India

- Delivered foundational concepts of machine learning to working professionals enrolled in the e-Masters program. Assisted with course content, coding assignments, and live Q&A sessions to clarify ML intuition and applications.

Teaching Assistant

IIT Kanpur

06/2022 – 09/2023   Kanpur, India

- Delivered foundational concepts of machine learning to working professionals enrolled in the e-Masters program. Assisted with course content, coding assignments, and live Q&A sessions to clarify ML intuition and applications.

Invited Speaker

Queen Mary University of London

London, United Kingdom

https://www.youtube.com/watch?v=l-0r2_q7Xxs

RESEARCH EXPERIENCE

- **Bridged a key research gap in domain adaptation for melody estimation** by introducing meta-learning techniques (eg, MAML) to MIR. Published supervised and confidence-based adaptation strategies in CODS-COMAD, MLSP and TASLP.
- **Designed and developed a CPU-friendly annotation tool** for singing melody extraction from polyphonic audio, significantly reducing manual labeling time and efforts. Built with Flask and HTML/CSS/JS. Currently under *double-blind* review.
- **Proposed a novel regression-based approach for melody estimation with uncertainty quantification**, addressing ambiguity in confidence prediction from the traditionally framed classification problem.
- **Proposed a novel nonconformity score** for conformal prediction to enable reliable post-hoc calibration of epistemic uncertainty, leveraging an Epinet-based Bayesian framework to address ambiguity in confidence estimation. Currently *in preparation*.
- **Identified gaps in the evaluation of speech LLMs for Indian languages and led the development of IndicST**, a multilingual synthetic dataset for AST tasks. Enabled benchmarking of models such as Whisper-v2 and SALMONN across resource-scarce Indian languages. Published the dataset in ICASSPW.

PUBLICATIONS

- **Kavya Ranjan Saxena**, Vipul Arora, 'Tutorial on understanding meta-learning for fast adaptation', CODS-COMAD 2023.
  https://dl.acm.org/doi/pdf/10.1145/3570991.3571030
- **Kavya Ranjan Saxena**, Vipul Arora, 'Meta-Learning-Based Supervised Domain Adaptation for Melody Extraction', 2024 IEEE MLSP, London, UK, 2024.
  https://ieeexplore.ieee.org/document/10734810
- **Kavya Ranjan Saxena**, Vipul Arora, 'Interactive Singing Melody Extraction Based on Active Adaptation', IEEE/ACM Transactions on Audio, Speech, and Language Processing (TASLP), 2024.
  https://ieeexplore.ieee.org/document/10530096
- **Kavya Ranjan Saxena**, Vipul Arora. 'Uncertainty Quantification in Melody Estimation using Histogram Representation'.
  https://arxiv.org/pdf/2505.05156
- Sanket Shah, **Kavya Ranjan Saxena**, Manideep Bharadwaj, Sharath Adavanne, Nagaraj Adiga, 'IndicST: Indian Multilingual Translation Corpus For Evaluating Speech Large Language Models', ICASSPW 2025.
  https://ieeexplore.ieee.org/document/11011192

SKILLS

Python, Tensorflow, PyTorch, LaTeX, GitHub Actions, HuggingFace(Transformers/Datasets), Technical Writing, Teaching/Mentoring

EDUCATION

IIT Kanpur   2019-2026

Ph.D. (CGPA: 8.68)

- **Discipline:** Signal Processing
- **Relevant Coursework:** Machine Learning for Signal Processing, Audio Representation Learning, Neural Networks

JIIT Noida   2012 - 2017

Dual Degree - B.Tech+M.Tech (CGPA: 7.5)

- **Discipline:** Electronics and Communications Engineering

SCHOLARSHIP/AWARDS

UGC-JRF Award Letter 2018